ASSIGNMENT 15.3

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Batch – 05

TASK – 01

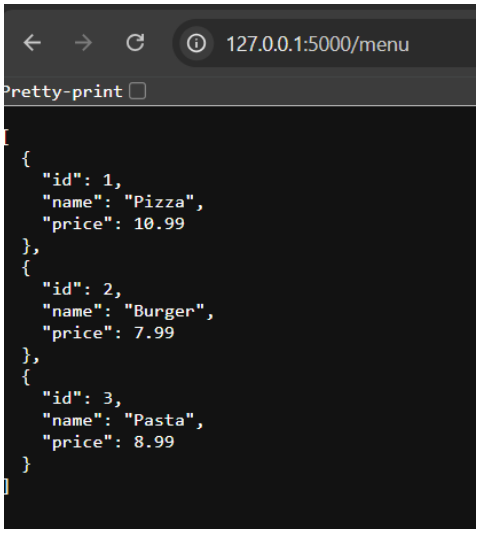
Prompt : Generate a API for online food ordering system in which we can get the menu, place an order,track the order, update existing order and cancel the order.

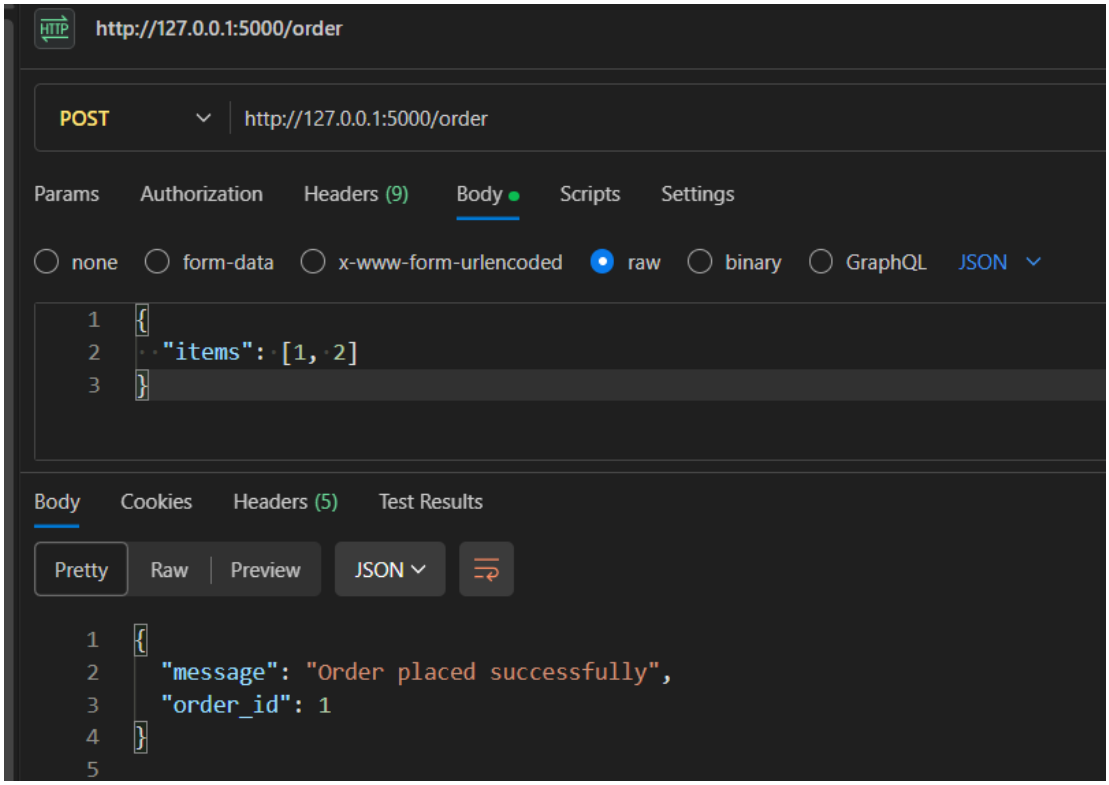
Code :

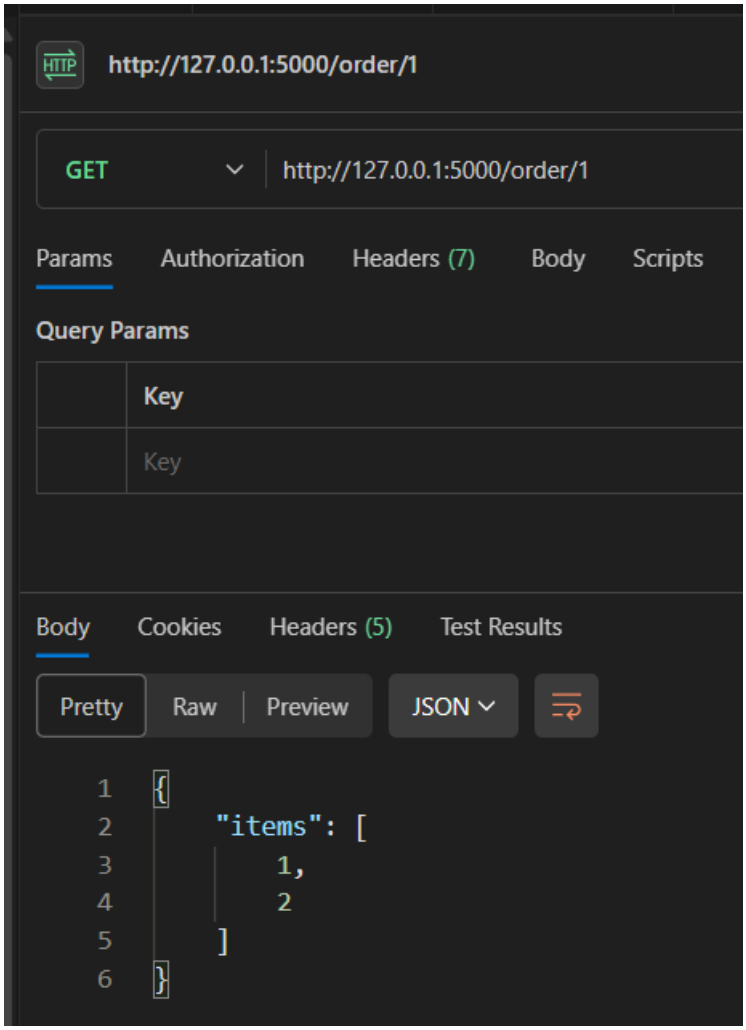
```
#Generate a API for online food ordering system in which we
#can get the menu, place an order,track the order, update existing order and cancel the order.
from flask import Flask, request, jsonify
app = Flask(__name__)
# Sample menu
menu = {
    1: {"name": "Pizza", "price": 10},
    2: {"name": "Burger", "price": 5},
    3: {"name": "Pasta", "price": 8}
}
# Sample orders
orders = {}
# Get the menu
@app.route('/menu', methods=['GET'])
def get_menu():
    return jsonify(menu)
# Place an order
@app.route('/order', methods=['POST'])
def place_order():
    order_data = request.get_json()
    order_id = len(orders) + 1
    orders[order_id] = order_data
    return jsonify({"message": "Order placed successfully", "order_id": order_id})
# Track the order
@app.route('/order/<int:order_id>', methods=['GET'])
def track_order(order_id):
    order = orders.get(order_id)
    if order:
        return jsonify(order)
    else:
        return jsonify({"message": "Order not found"}), 404
# Update existing order
@app.route('/order/<int:order_id>', methods=['PUT'])
```

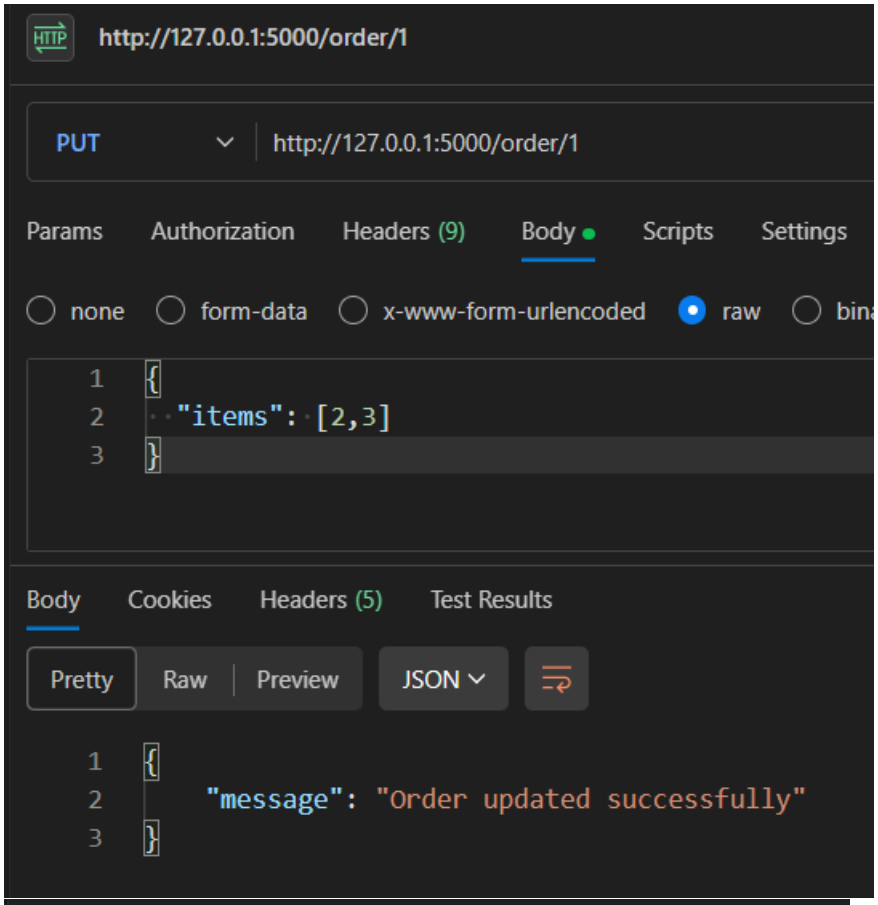
```
def update_order(order_id):
    order_data = request.get_json()
    if order_id in orders:
        orders[order_id] = order_data
        return jsonify({"message": "Order updated successfully"})
    else:
        return jsonify({"message": "Order not found"}), 404
# Cancel the order
@app.route('/order/<int:order_id>', methods=['DELETE'])
def cancel_order(order_id):
    if order_id in orders:
        del orders[order_id]
        return jsonify({"message": "Order cancelled successfully"})
    else:
        return jsonify({"message": "Order not found"}), 404
if __name__ == '__main__':
    app.run(debug=True)
# Explanation:
# In this code, we have created a simple API for an online food ordering system using Flask.
# The API has endpoints to get the menu, place an order, track an order, update an existing order, and cancel an order.
# The menu is represented as a dictionary, and orders are stored in another dictionary with unique order IDs.
# Each endpoint handles the respective HTTP methods (GET, POST, PUT, DELETE) to perform the required operations on the orders.
```

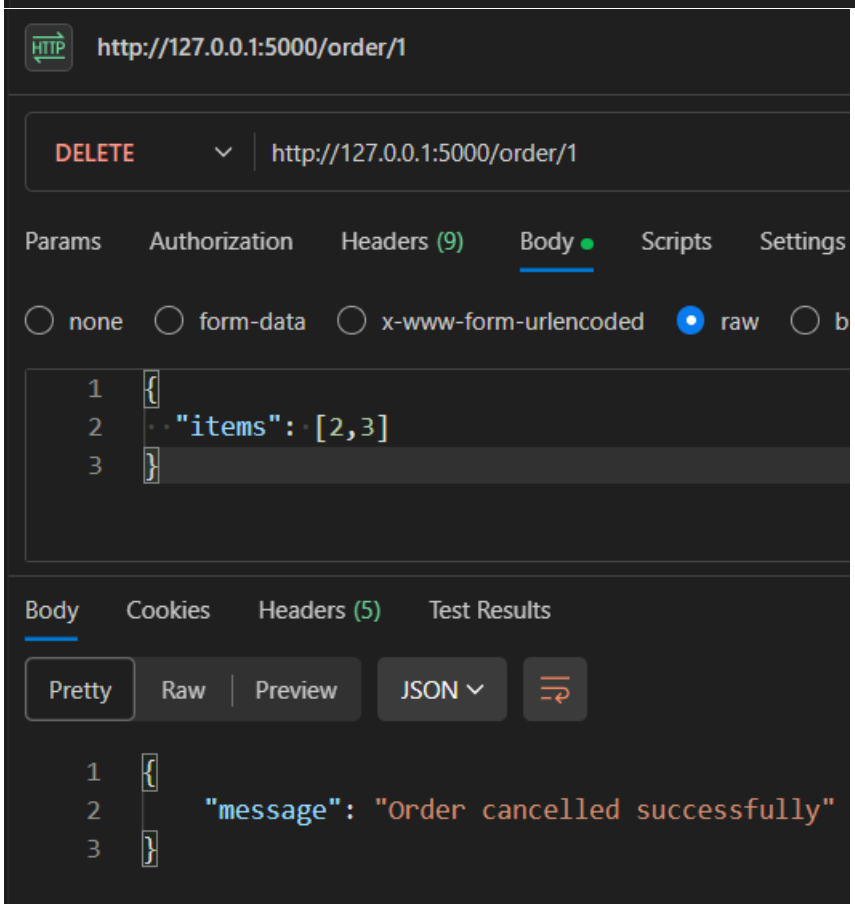
Output:




http://127.0.0.1:5000/order
POST
http://127.0.0.1:5000/order
Params
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
{
"items": [1, 2]
}
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"message": "Order placed successfully",
"order_id": 1
}


http://127.0.0.1:5000/order/1
GET
http://127.0.0.1:5000/order/1
Params
Authorization
Headers (7)
Body
Scripts
Query Params
Key
Key
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"items": [
1,
2
]
}


http://127.0.0.1:5000/order/1
PUT
http://127.0.0.1:5000/order/1
Params
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
{
"items": [2,3]
}
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"message": "Order updated successfully"
}


http://127.0.0.1:5000/order/1
DELETE
http://127.0.0.1:5000/order/1
Params
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
{
"items": [2,3]
}
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"message": "Order cancelled successfully"
}

Explanation : This Flask-based code creates a basic API for an online food ordering system. It lets users check the available menu, place new orders, view or track existing orders, modify them, and also cancel orders if needed. The application uses Python dictionaries to store menu items and order details. Different HTTP methods (like GET, POST, PUT, DELETE) are used to handle these actions, allowing the API to manage orders efficiently.

TASK – 02 :

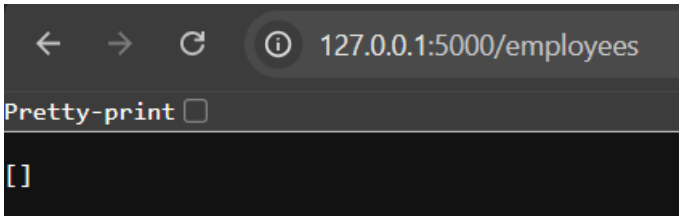
Prompt : Generate a API for Employee Payroll System in which we can list all employees, add a new employee, update an existing employee's salary, and delete an employee.

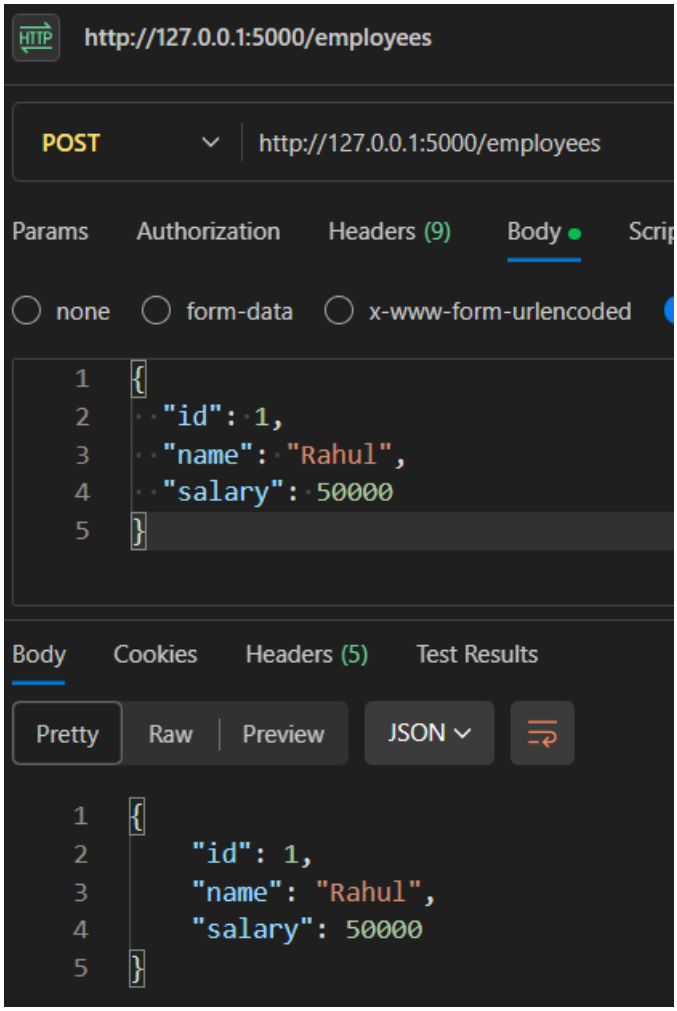
Code :

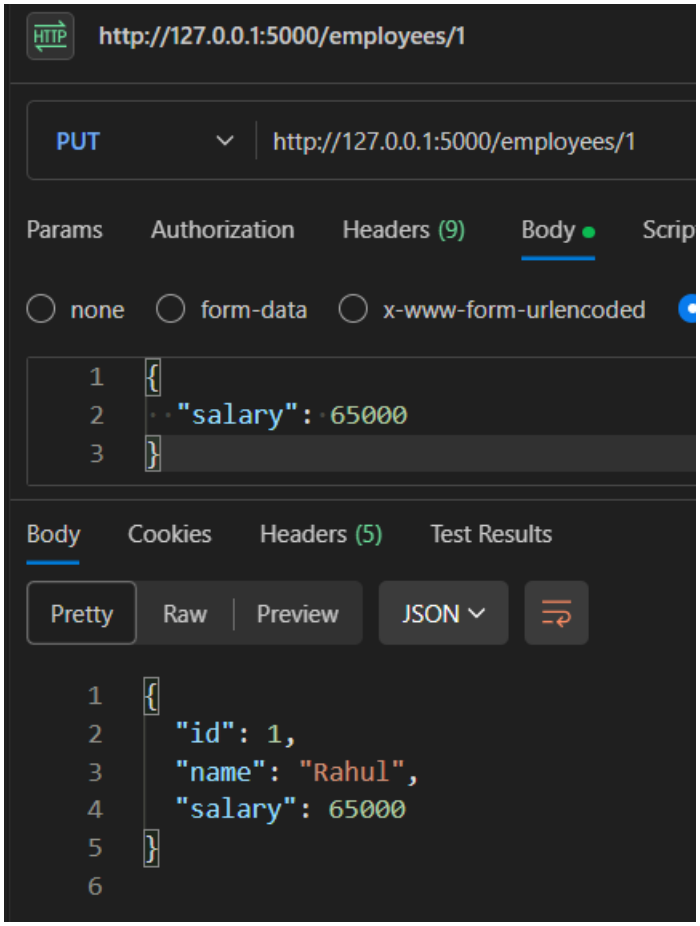
```
#Generate a API for Employee Payroll System in which we can list all
employees, add a new employee, update an existing employee's salary, and
delete an employee.
from flask import Flask, request, jsonify
app = Flask(__name__)
# In-memory database for employees
employees = []
# API to list all employees
@app.route('/employees', methods=['GET'])
def list_employees():
    return jsonify(employees)
# API to add a new employee
@app.route('/employees', methods=['POST'])
def add_employee():
    new_employee = request.get_json()
    employees.append(new_employee)
    return jsonify(new_employee), 201
# API to update an existing employee's salary
@app.route('/employees/<int:employee_id>', methods=['PUT'])
def update_employee_salary(employee_id):
    updated_data = request.get_json()
    for employee in employees:
        if employee['id'] == employee_id:
            employee['salary'] = updated_data['salary']
            return jsonify(employee)
    return jsonify({'error': 'Employee not found'}), 404
# API to delete an employee
@app.route('/employees/<int:employee_id>', methods=['DELETE'])
def delete_employee(employee_id):
    global employees
```

```
    employees = [employee for employee in employees if employee['id'] !=
employee_id]
    return jsonify({'message': 'Employee deleted'})
if __name__ == '__main__':
    app.run(debug=True)
# Explanation:
# This code defines a simple API for an Employee Payroll System using Flask.
It includes endpoints to list all employees, add a new employee, update an
existing employee's salary, and delete an employee. The employee data is
stored in an in-memory list for simplicity. Each endpoint handles the
corresponding HTTP method (GET, POST, PUT, DELETE) to perform the required
operations on the employee data.
```

Output :




http://127.0.0.1:5000/employees
POST
http://127.0.0.1:5000/employees
Params
Authorization
Headers (9)
Body
Scrip
none
form-data
x-www-form-urlencoded
{
"id": 1,
"name": "Rahul",
"salary": 50000
}
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"id": 1,
"name": "Rahul",
"salary": 50000
}



Explanation : This Flask-based Employee Payroll API provides RESTful endpoints to handle employee records. It enables users to retrieve a list of employees, create new employee entries, modify existing salaries, and remove employees when needed. The system uses an in-memory list to store employee details, and all responses are delivered in JSON format.

TASK – 03

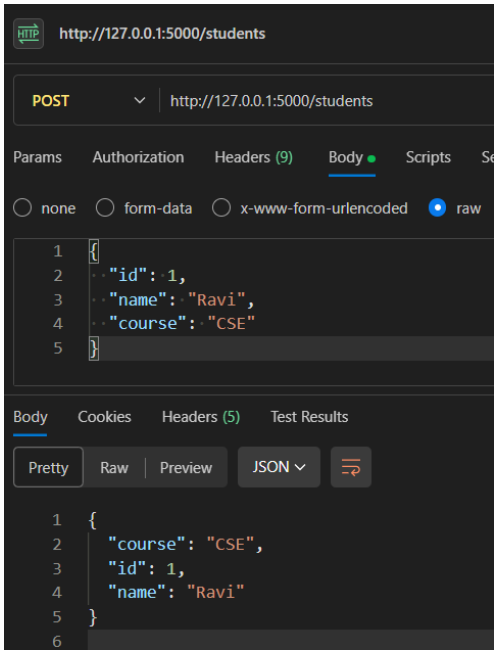
Prompt : Generate a API for Student Information System in which we can list all students, add a new student, update an existing student's information, and delete a student.

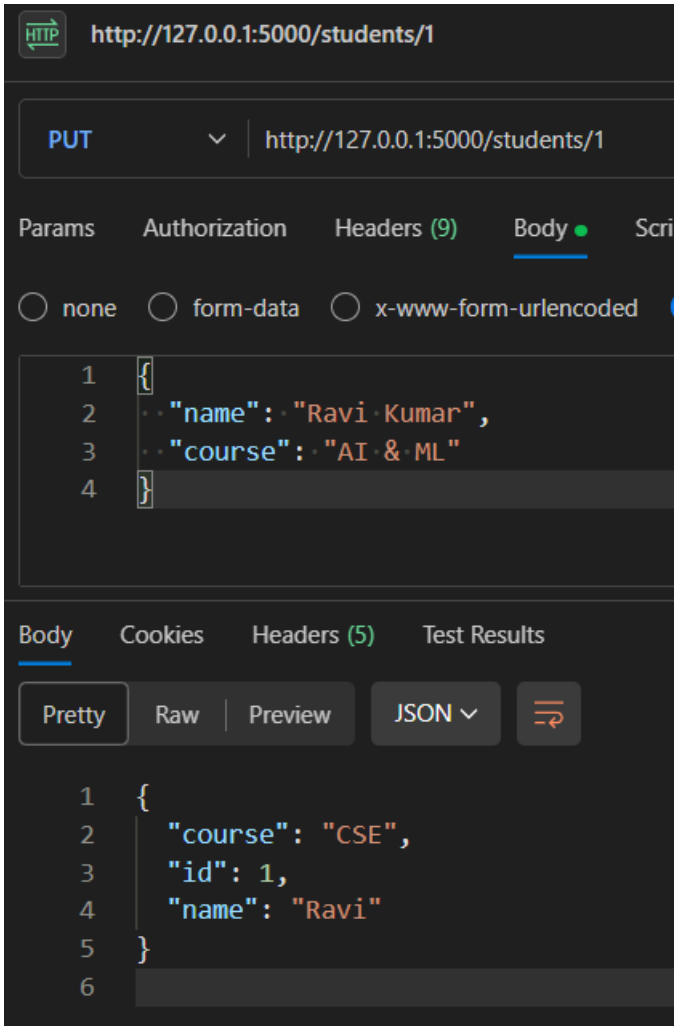
Code :

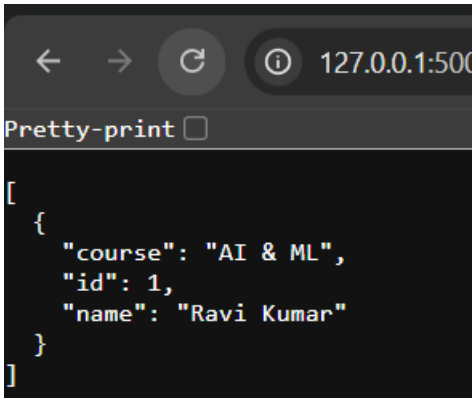
```
#Generate a API for Student Information System in which we can list all
students, add a new student, update an existing student's information, and
delete a student.
from flask import Flask, jsonify, request
app = Flask(__name__)
students = []
@app.route('/students', methods=['GET'])
```

```python
def get_students():
    return jsonify(students)
@app.route('/students', methods=['POST'])
def add_student():
    new_student = request.get_json()
    students.append(new_student)
    return jsonify(new_student), 201
@app.route('/students/<int:id>', methods=['PUT'])
def update_student(id):
    updated_student = request.get_json()
    for student in students:
        if student['id'] == id:
            student.update(updated_student)
            return jsonify(student)
    return jsonify({'message': 'Student not found'}), 404
@app.route('/students/<int:id>', methods=['DELETE'])
def delete_student(id):
    for student in students:
        if student['id'] == id:
            students.remove(student)
            return jsonify({'message': 'Student deleted'})
    return jsonify({'message': 'Student not found'}), 404
if __name__ == '__main__':
    app.run(debug=True)
```

Output :







Explanation : This Flask application implements a basic API for managing student information. It supports core CRUD operations, allowing users to retrieve all student records, add new students, modify existing student details, and remove students using their ID. The student data is maintained in a list named students, which acts as the in-memory storage for the system.

TASK – 04

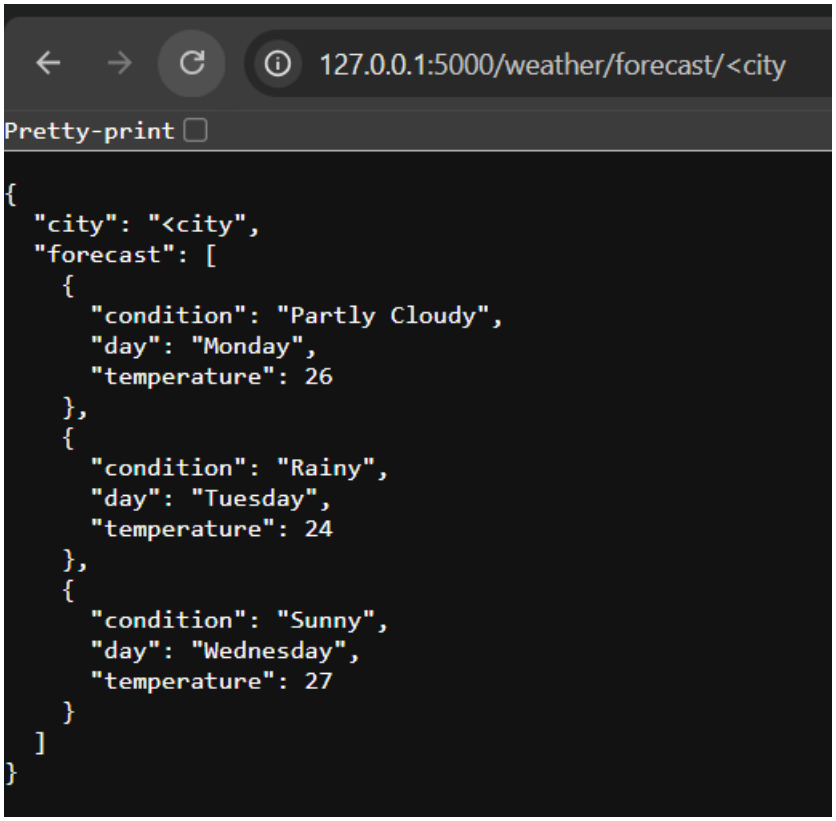
Prompt : Create a Python Flask REST API that integrates with the OpenWeatherMap API to fetch weather data. Add endpoints /weather/current/<city> and /weather/forecast/<city> that return JSON with temperature, humidity, and weather condition.

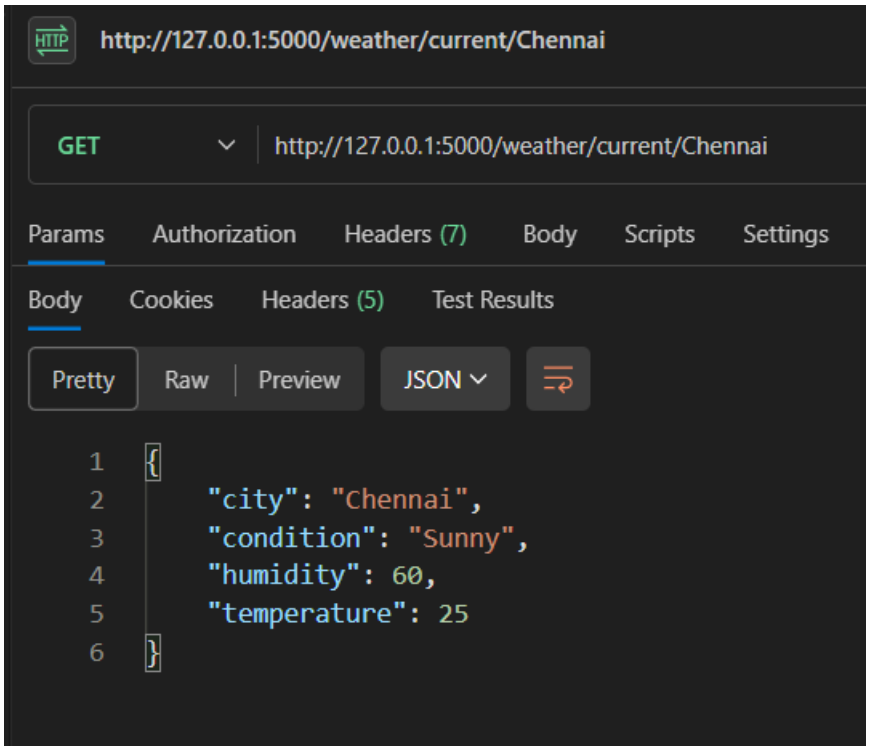
Code :

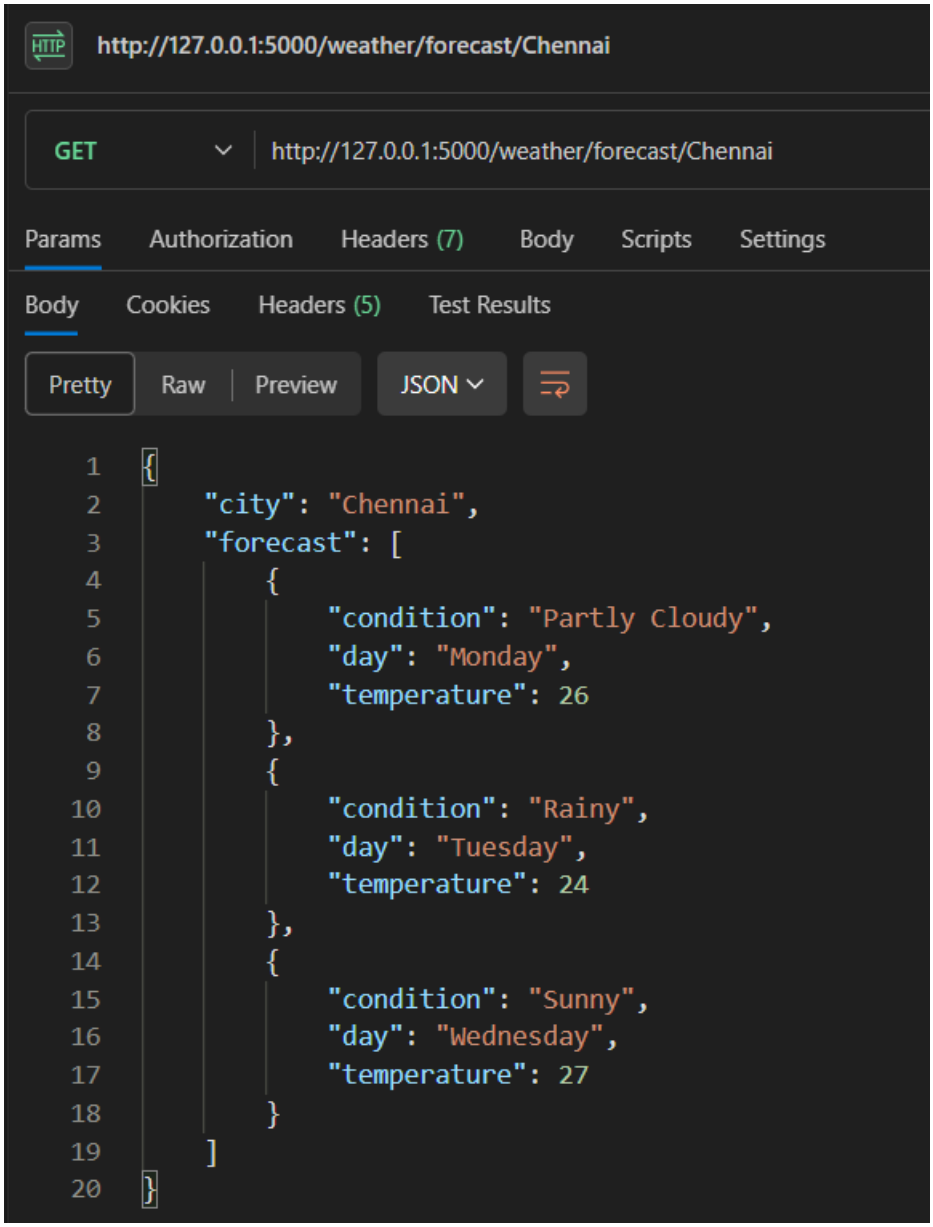
```
#Create a Python Flask REST API that integrates with the OpenWeatherMap API to
fetch weather data.
#Add endpoints /weather/current/<city> and /weather/forecast/<city> that
return JSON with temperature, humidity, and weather condition.
from flask import Flask, jsonify

app = Flask(__name__)
@app.route('/weather/current/<city>', methods=['GET'])
def get_current_weather(city):
    # Here you would integrate with the OpenWeatherMap API to fetch current
weather data for the specified city
    # For demonstration purposes, we'll return a mock response
    weather_data = {
        "city": city,
        "temperature": 25,
        "humidity": 60,
        "condition": "Sunny"
    }
    return jsonify(weather_data)
@app.route('/weather/forecast/<city>', methods=['GET'])
def get_weather_forecast(city):
    # Here you would integrate with the OpenWeatherMap API to fetch weather
forecast data for the specified city
    # For demonstration purposes, we'll return a mock response
    forecast_data = {
        "city": city,
        "forecast": [
            {"day": "Monday", "temperature": 26, "condition": "Partly
Cloudy"},
            {"day": "Tuesday", "temperature": 24, "condition": "Rainy"},
            {"day": "Wednesday", "temperature": 27, "condition": "Sunny"}
        ]
    }
    return jsonify(forecast_data)
if __name__ == '__main__':
    app.run(debug=True)
```

Output :


127.0.0.1:5000/weather/forecast/<city
Pretty-print
{
"city": "<city",
"forecast": [
{
"condition": "Partly Cloudy",
"day": "Monday",
"temperature": 26
},
{
"condition": "Rainy",
"day": "Tuesday",
"temperature": 24
},
{
"condition": "Sunny",
"day": "Wednesday",
"temperature": 27
}
]
}


http://127.0.0.1:5000/weather/current/Chennai
GET
http://127.0.0.1:5000/weather/current/Chennai
Params
Authorization
Headers (7)
Body
Scripts
Settings
Body
Cookies
Headers (5)
Test Results
Pretty
Raw
Preview
JSON
{
"city": "Chennai",
"condition": "Sunny",
"humidity": 60,
"temperature": 25
}



Explanation : This Flask-based code defines a simple REST API with two routes: /weather/current/<city> and /weather/forecast/<city>. The first route provides sample current weather details for a given city, while the second returns a sample forecast for the upcoming days. In a real-world scenario, the placeholder data would be replaced by live weather information fetched from an external service like the OpenWeatherMap API.

TASK – 05

Prompt : Create a Python **Flask REST API endpoint to delete an item from a list using the DELETE method. The endpoint should be /items/<int:index>. If the index is invalid, return a JSON error message "Item not found" with status

code 404. If successful, remove the item and return a JSON response with "message": "Item deleted" and the deleted item.

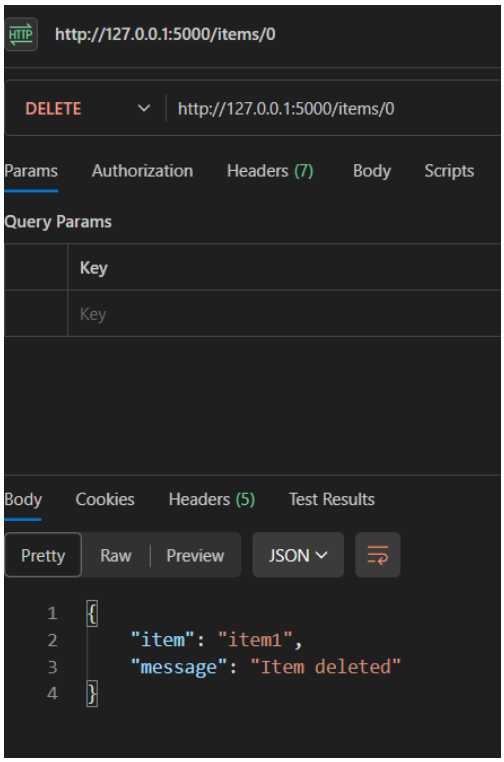
Code :

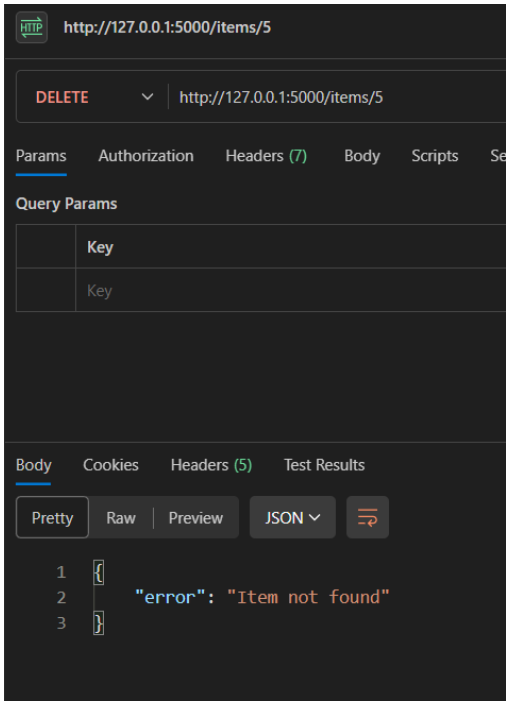
```
#Create a Python **Flask REST API endpoint to delete an item from a list using the DELETE method. The endpoint should be /items/<int:index>. If the index is invalid, return a JSON error message "Item not found" with status code 404. If successful, remove the item and return a JSON response with "message": "Item deleted" and the deleted item.

from flask import Flask, jsonify, request
app = Flask(__name__)
items = ["item1", "item2", "item3"]

@app.route('/items/<int:index>', methods=['DELETE'])
def delete_item(index):
    if 0 <= index < len(items):
        deleted_item = items.pop(index)
        return jsonify({"message": "Item deleted", "item": deleted_item})
    else:
        return jsonify({"error": "Item not found"}), 404
if __name__ == '__main__':
    app.run(debug=True)
```

Output :





Explanation :

This Flask code implements a REST API endpoint for removing an item from a list. The route /items/<int:index> accepts an index value that specifies which item should be deleted. When a DELETE request is sent to this endpoint, the delete_item function is triggered with the given index. It first verifies whether the index is within the valid range of the list. If the index is valid, the function deletes the item using the pop() method and returns a JSON response confirming the deletion, along with the removed item.